

Exercise lecture 2

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March 2026

Let $X \sim N(0, \sigma^2)$. To show:

$$\mathbb{E}[X^4] = 3\sigma^4.$$

Let $Z \sim N(0, 1)$. Then

$$X = \sigma Z \Rightarrow X^4 = \sigma^4 Z^4.$$

$$\Rightarrow \mathbb{E}[X^4] = \sigma^4 \mathbb{E}[Z^4].$$

We have

$$\mathbb{E}[Z^4] = \int_{-\infty}^{\infty} z^4 \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} z^4 e^{-z^2/2} dz.$$

Using integration by parts:

let

$$I = \int_0^{\infty} z^4 e^{-z^2/2} dz = \int_0^{\infty} z^3 \cdot z e^{-z^2/2} dz.$$

Let

$$u = z^3, \quad dv = z e^{-z^2/2} dz \Rightarrow du = 3z^2 dz, \quad v = -e^{-z^2/2}.$$

$$\Rightarrow I = \left[-z^3 e^{-z^2/2} \right]_0^{\infty} + \int_0^{\infty} 3z^2 e^{-z^2/2} dz = 3 \int_0^{\infty} z^2 e^{-z^2/2} dz.$$

Use integration by parts again:

let

$$u = z, \quad dv = z e^{-z^2/2} dz \Rightarrow du = dz, \quad v = -e^{-z^2/2}.$$

$$\Rightarrow \int_0^{\infty} z^2 e^{-z^2/2} dz = \left[-z e^{-z^2/2} \right]_0^{\infty} + \int_0^{\infty} e^{-z^2/2} dz = \int_0^{\infty} e^{-z^2/2} dz.$$

We use the known Gaussian integral:

$$\int_{-\infty}^{\infty} e^{-z^2/2} dz = \sqrt{2\pi}.$$

Thus,

$$\int_0^{\infty} e^{-z^2/2} dz = \frac{\sqrt{2\pi}}{2}.$$

Therefore,

$$I = 3 \cdot \frac{\sqrt{2\pi}}{2}.$$

So we get

$$\mathbb{E}[Z^4] = \frac{2}{\sqrt{2\pi}} \cdot I = \frac{2}{\sqrt{2\pi}} \cdot \left(3 \frac{\sqrt{2\pi}}{2} \right) = 3.$$

And thus

$$\mathbb{E}[X^4] = \sigma^4 \mathbb{E}[Z^4] = 3\sigma^4.$$